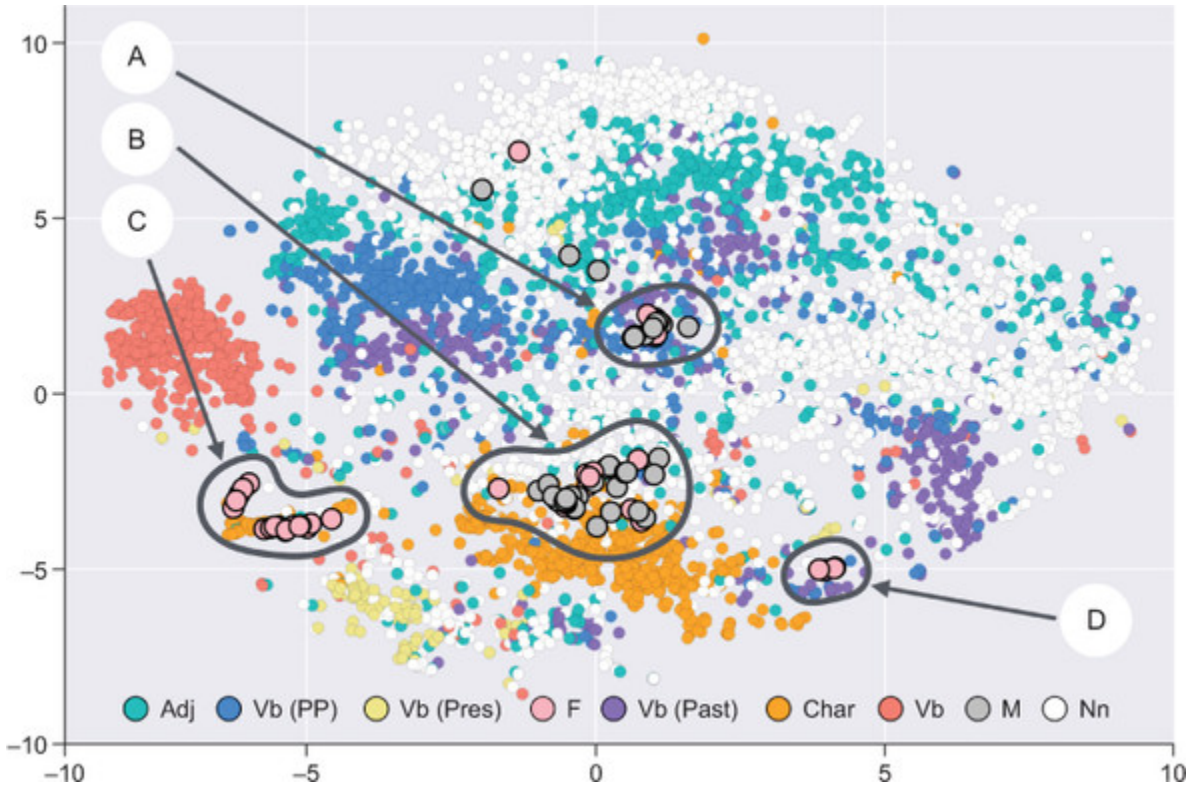


In control tasks, we similarly have a space of data to process, but each piece of data also has a time dimension—the data exists in both time and space. This means that what the algorithm decides at one time is influenced by what happened at a previous time. This isn't the case for ordinary image classification and similar problems. Time makes the training task dynamic—the data set upon which the algorithm is training is not necessarily fixed but changes based on the decisions the algorithm makes.

Ordinary image classification-like tasks fall under the category of *supervised learning*, because the algorithm is trained on how to properly classify images by giving it the right answers. The algorithm at first takes random guesses, and it is iteratively corrected until it learns the features in the image that correspond to the appropriate label. This requires us to already know what the right answers are, which can be cumbersome. If you want to train a deep learning algorithm to correctly classify images of various species of plants, you would have to painstakingly acquire thousands of such images and manually associate class labels with each one and prepare the data in a format that a machine learning algorithm can operate on, generally some type of matrix.